



FORM TP 2011022

JANUARY 2011

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of **SIX** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.
6. Mathematical tables are provided.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. A popular sprinter of mass 60 kg, was running a 100 m race. Her velocity was measured over a 10.0 s period. The results are recorded in Table 1.

TABLE 1

Velocity, v/ms^{-1}	0.0	2.5	5.0	7.5	10.0	10.0
Time, t/s	0.0	2.0	4.0	6.0	8.0	10.0

- (a) Define the following terms:

Velocity

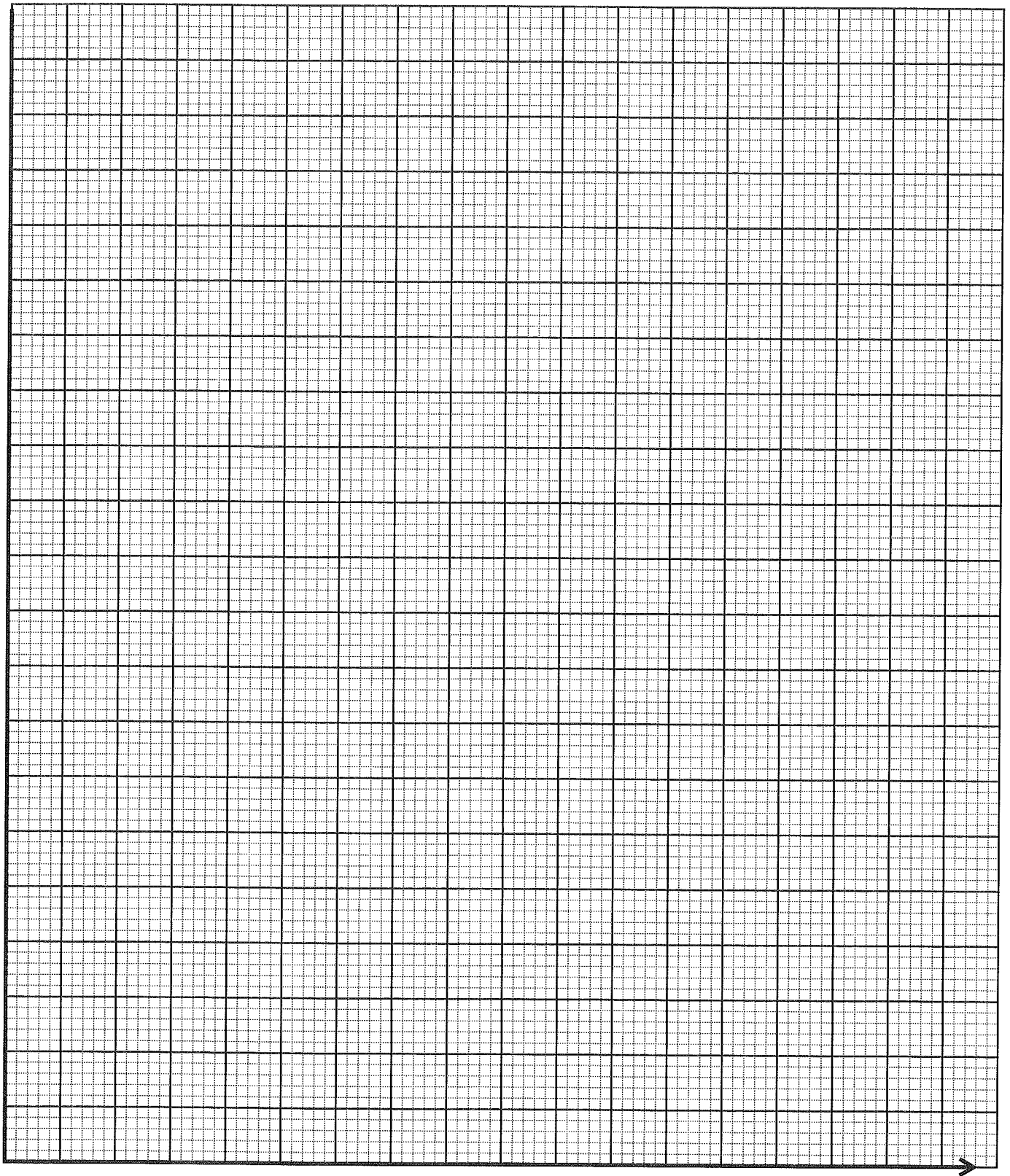
(2 marks)

Acceleration

(2 marks)

- (b) Use the results from Table 1 to plot a graph of Velocity versus Time on page 3.
(7 marks)
- (c) Determine the slope of the graph over the first 6.0 s of the race.

(4 marks)



- (d) How is the slope of the graph related to the acceleration of the sprinter?

(2 marks)

- (e) Calculate the resultant force acting on the sprinter after 6.0 s.

(3 marks)

- (f) Using the graph drawn on page 3, determine how far from the finish line the sprinter would be after 10.0 s.

(5 marks)

Total 25 marks

2. Electromagnetic waves consist of visible light and radiations higher or lower than the wavelength of light.

(a) (i) Identify ONE type of radiation with wavelength:

a) longer than visible light _____

b) shorter than visible light. _____

(2 marks)

(ii) Complete Table 2 which relates to the sources and uses of electromagnetic waves.

TABLE 2

Name of Wave	Source	Use
X-ray		To take X-ray pictures
Gamma Ray		
Radio wave		

(5 marks)

(b) (i) If gamma rays have a wavelength of 3.0×10^{-12} m, calculate the frequency of this type of radiation.

[Velocity of all electromagnetic waves = $3.0 \times 10^8 \text{ m s}^{-1}$]

(3 marks)

- (ii) A ray of red light emerges from a glass block as shown in Figure 1.

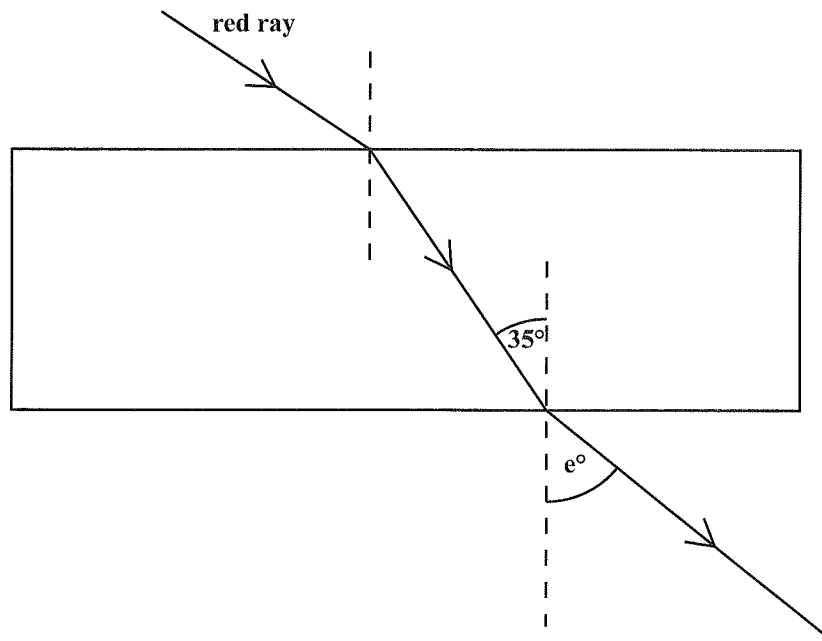


Figure 1

If the block has a refractive index of 1.5, determine the value of the angle e .

(5 marks)

Total 15 marks

3. (a) Complete Table 3 which relates electrical circuit symbols to the names of electrical components.

TABLE 3

Circuit Symbol	Name of Components
	Variable resistor
	Semi-conductor diode

(6 marks)

- (b) Figure 2 shows a circuit diagram.

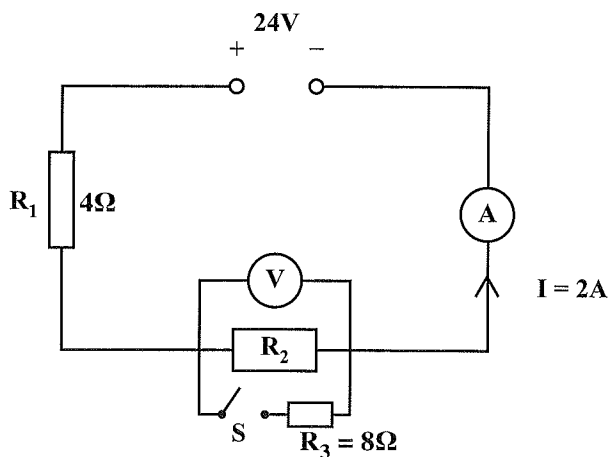


Figure 2

In the circuit shown in Figure 2, $R_1 = 4\Omega$, $R_3 = 8\Omega$ and the supply is 24V. With switch S open, the current I through the ammeter is 2 A.

- (i) State the formula which relates voltage, V, current, I and resistance, R.

(1 mark)

- (ii) Calculate the value of R_2 .

(3 marks)

- (iii) What is the reading on voltmeter $\textcircled{V}$?

(2 marks)

- (iv) If switch S is closed, what would be the new reading on ammeter, $\textcircled{A}$?

(3 marks)

Total 15 marks

SECTION B

Attempt ALL questions.

You MUST write your answers in the space provided after each question.

4. (a) Describe how the apparatus shown in Figure 3 below may be used to show how the temperature of a substance such as naphthalene, varies during cooling. Include in your description any other equipment that may be necessary.

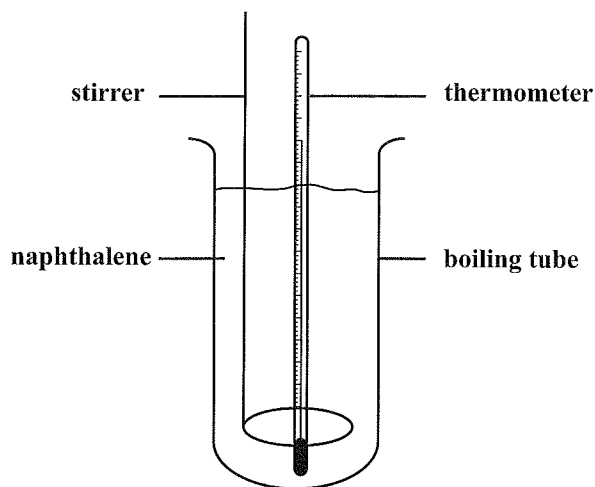


Figure 3

(6 marks)

- (b) In an experiment to determine the specific latent heat of ice, the following results were obtained.

Initial mass of water at 30°C = 100 g

Final mass of water (initial water plus melted ice) at 20°C = 110 g

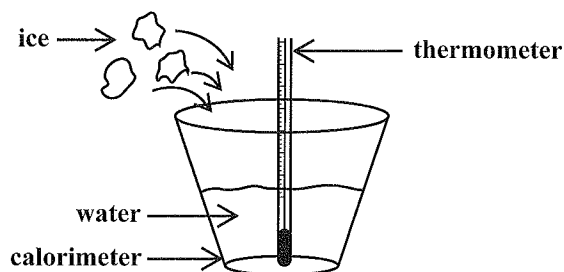


Figure 4

Assume the container has negligible heat capacity and that the heat lost to the surroundings is equal to the heat gained from the surroundings during the experiment.

GO ON TO THE NEXT PAGE

- (i) Determine the mass of the melted ice. (1 mark)
- (ii) Calculate the heat lost by the water. (3 marks)
- (iii) Write down an expression for the TOTAL heat gained by the ice in melting and the melted ice in warming to 20°C. (2 marks)
- (iv) Hence calculate the specific latent heat of fusion of ice. (3 marks)

[Specific Heat Capacity of water = $4.2 \text{ J g}^{-1} \text{ K}^{-1}$]

Total 15 marks

Write your answer to Question 4 here.

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5. (a) With the aid of at least one diagram, describe how the magnitude of the electromotive force (e.m.f.) induced in a conductor depends on the rate of change of magnetic flux that the conductor experiences. (6 marks)
- (b) To make electric cars more efficient, they will need to be charged regularly using a magnetic field below the road surface. One solution is to designate one lane of the highway for recharging batteries.
- (i) In the charging lane cars are required to keep moving. Why should this be so? (3 marks)
- (ii) Assuming the magnetic field is constant, state with reason, how the speed of the car will affect the charging of the batteries. (3 marks)
- (iii) In a certain section of road, 200 KW of power is expended to charge 100 cars. If each car converts 600 Watts of power to charging the battery, calculate the conversion efficiency. (3 marks)

Total 15 marks

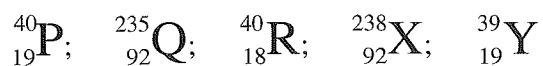
Write your answer to Question 5 here.

6. (a) Some great scientists contributed to the modern-day view of the model of the atom. Two of these were J. J. Thompson and Ernest Rutherford.

For EACH of these scientists, describe briefly TWO main ideas they put forward concerning the nature of the atom, including the name of EACH of their models.

(6 marks)

- (b) In final revision for your CSEC Physics examinations, your teacher gave you the following data on some unknown elements:



You are required to determine:

- (i) which nuclides have the identical mass number
- (ii) how many neutrons are in the heaviest nuclide
- (iii) which nuclides are isotopes.

(5 marks)

- (c) In a half-life experiment, the data shown in Table 4 were obtained.

TABLE 4

Count rate / minute	Time / second
4000	0.0
1000	55.0
500	80.0

Use the data to determine:

- (i) TWO different values of the half-life of the element
- (ii) The average half-life of the element

(4 marks)

Total 15 marks

Write your answer to Question 6 here.

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